

Catalog Storefront — Event Storming Canvas

Subdomain: Catalog Storefront **Format:** Big Picture Event Storming **Status:** Draft

Reading Guide

- **Domain Events** (orange) — Things that happened, past tense
 - **Commands** (blue) — Actions that trigger events
 - **Aggregates** (yellow) — Entities that process commands and emit events
 - **Actors** (small yellow) — Who issues the command
 - **Read Models** (green) — Views/projections consumed by actors
 - **External Systems** (pink) — Systems outside this domain
 - **Policies** (lilac) — “Whenever X happens, then Y” automation rules
 - **Pivotal Events** (orange with red border) — Events that mark phase transitions or boundary shifts
-

Visual Timeline

```
timeline
    title Catalog Storefront – Event Storming Timeline

    section Catalog Management (upstream)
        Product Added to Catalog 📌🌟 : Product Specifications Updated
        Product Price Changed : Product Availability Changed
        Product Discontinued 📌🌟 : Technical Document Uploaded
        Category Created : Category Hierarchy Updated
        Product Categorized

    section Product Discovery (JS1)
        Catalog Searched : Search Filters Applied
        Search Results Sorted : Search Page Retrieved
        Suggestions Retrieved

    section Product Evaluation (JS2)
        Product Details Viewed : Technical Document Downloaded
        Product Pricing Viewed : Product Availability Checked
        Product Reviews Viewed : Compatible Products Retrieved
        Similar Products Retrieved : Frequently Bought Together Retrieved

    section Product Comparison (JS3)
        Product Added to Comparison : Product Removed from Comparison
        Comparison Matrix Generated 📌🌟

    section Compatibility Verification (JS4)
        Compatibility Check Requested : Voltage Compatibility Evaluated
```

Current Rating Evaluated : Physical Fit Evaluated
Protocol Evaluated : Weight/Thrust Evaluated
Compatibility Check Completed   : Compatible Replacements Retrieved

section Catalog Browsing (JS5)
Category Tree Retrieved : Category Browsed
Category Facets Retrieved : Category Products Filtered

section Partner Catalog Access (JS6)
Catalog Feed Generated   : Batch Availability Retrieved
Batch Pricing Retrieved

Legend:  = Domain Event |  = Pivotal Event (marks phase transition or boundary shift)

Timeline: Left → Right

1. Catalog Management (upstream — feeds our domain)

[External: Catalog Admin]

Command: Add Product to Catalog
Aggregate: Product
→ Event: Product Added to Catalog  PIVOTAL

Command: Update Product Specifications
Aggregate: Product
→ Event: Product Specifications Updated

Command: Update Product Pricing
Aggregate: Product
→ Event: Product Price Changed

Command: Upload Technical Document
Aggregate: Technical Document
→ Event: Technical Document Uploaded

Command: Discontinue Product
Aggregate: Product
→ Event: Product Discontinued  PIVOTAL

Command: Restock Product
Aggregate: Product
→ Event: Product Availability Changed

```
Command: Create Category
Aggregate: Category
  → Event: Category Created

Command: Reorganize Category Hierarchy
Aggregate: Category
  → Event: Category Hierarchy Updated

Command: Assign Product to Category
Aggregate: Product
  → Event: Product Categorized
```

Policy: Whenever Product Added to Catalog → Index Product for Search **Policy:** Whenever Product Specifications Updated → Reindex Product for Search **Policy:** Whenever Product Specifications Updated → Update Compatibility Rules Cache **Policy:** Whenever Product Discontinued → Remove from Active Search Index **Policy:** Whenever Product Price Changed → Notify Partner Feed Cache

2. Product Discovery (JS1)

```
[Actor: Professional Integrator / Fleet Buyer / Casual Browser]
[Read Model: Search Results]

Command: Search Catalog
Aggregate: Search Index
  → Event: Catalog Searched

Command: Apply Specification Filters
Aggregate: Search Index
  → Event: Search Filters Applied

Command: Sort Search Results
Aggregate: Search Index
  → Event: Search Results Sorted

Command: Request Next Page
Aggregate: Search Index
  → Event: Search Page Retrieved

Command: Request Autocomplete Suggestions
Aggregate: Search Index
  → Event: Suggestions Retrieved
```

Read Model: Search Results (query, filters, sorted product summaries, pagination) **Read Model:** Autocomplete Suggestions (terms, categories, products)

3. Product Evaluation (JS2)

[Actor: Professional Integrator / Fleet Buyer]
[Read Model: Product Detail View]

Command: View Product Details
Aggregate: Product
→ Event: Product Details Viewed

Command: Download Technical Document
Aggregate: Technical Document
→ Event: Technical Document Downloaded

Command: Check Product Pricing
Aggregate: Product
→ Event: Product Pricing Viewed

Command: Check Product Availability
Aggregate: Product
→ Event: Product Availability Checked

Command: View Product Reviews
Aggregate: Review
→ Event: Product Reviews Viewed

Command: Browse Compatible Products
Aggregate: Product
→ Event: Compatible Products Retrieved

Command: Browse Similar Alternatives
Aggregate: Product
→ Event: Similar Products Retrieved

Command: Browse Frequently Bought Together
Aggregate: Product
→ Event: Frequently Bought Together Retrieved

Read Model: Product Detail (full specs, images, pricing, availability, documents) **Read Model:** Related Products (compatible, similar, frequently bought together)

Policy: Whenever Product Details Viewed → Track View for Analytics **Policy:** Whenever Technical Document Downloaded → Log Download for Compliance Audit

4. Product Comparison (JS3)

[Actor: Professional Integrator / Fleet Buyer]
[Read Model: Comparison Matrix]

```
Command: Add Product to Comparison
Aggregate: Comparison
  → Event: Product Added to Comparison

Command: Remove Product from Comparison
Aggregate: Comparison
  → Event: Product Removed from Comparison

Command: Generate Comparison Matrix
Aggregate: Comparison
  → Event: Comparison Matrix Generated ★ PIVOTAL
```

Read Model: Comparison Matrix (shared specs, differing specs, side-by-side values)

Note: The Comparison Matrix Generated event is pivotal because it often directly precedes a purchase decision or a compatibility check — it marks the transition from “exploring” to “deciding.”

5. Compatibility Verification (JS4) — SEPARATE BOUNDED CONTEXT

```
[Actor: Professional Integrator / Fleet Buyer]
[Read Model: Compatibility Report]

Command: Submit Component Set for Verification
Aggregate: Compatibility Check
  → Event: Compatibility Check Requested

[Internal processing – rule engine evaluates:]
  → Event: Voltage Compatibility Evaluated
  → Event: Current Rating Compatibility Evaluated
  → Event: Physical Fit Compatibility Evaluated
  → Event: Protocol Compatibility Evaluated
  → Event: Weight/Thrust Compatibility Evaluated

  → Event: Compatibility Check Completed ★ PIVOTAL
    (status: pass | fail | partial)

Command: Request Compatible Replacements
Aggregate: Compatibility Check
  → Event: Compatible Replacements Retrieved
```

Read Model: Compatibility Report (pass/fail, incompatibility details, severity, replacement suggestions)

Policy: Whenever Compatibility Check Completed (fail) → Suggest Compatible Replacements

Policy: Whenever Compatibility Check Completed → Log Check for Analytics

Hotspot 🔥 : Compatibility rule management — who maintains the rules? How are new component types added? This is a separate concern from the check itself.

6. Catalog Browsing (JS5)

```
[Actor: Casual Browser / Professional Integrator]
[Read Model: Category Navigation]
```

```
Command: Request Category Tree
Aggregate: Category
  → Event: Category Tree Retrieved
```

```
Command: Browse Category
Aggregate: Category
  → Event: Category Browsed
```

```
Command: Request Category Facets
Aggregate: Category
  → Event: Category Facets Retrieved
```

```
Command: Filter Category Products
Aggregate: Category
  → Event: Category Products Filtered
```

Read Model: Category Navigation (hierarchy, product counts, available facets)

7. Partner Catalog Access (JS6) — SEPARATE BOUNDED CONTEXT

```
[Actor: Partner System]
[Read Model: Catalog Feed]
```

```
Command: Request Catalog Feed
Aggregate: Catalog Feed
  → Event: Catalog Feed Generated ★ PIVOTAL
```

```
Command: Request Batch Availability
Aggregate: Catalog Feed
  → Event: Batch Availability Retrieved
```

```
Command: Request Batch Pricing
Aggregate: Catalog Feed
  → Event: Batch Pricing Retrieved
```

Read Model: Partner Catalog Feed (simplified product projections, bulk data)

Policy: Whenever Product Added to Catalog → Update Partner Feed Cache **Policy:** Whenever Product Price Changed → Update Partner Feed Cache **Policy:** Whenever Product Availability

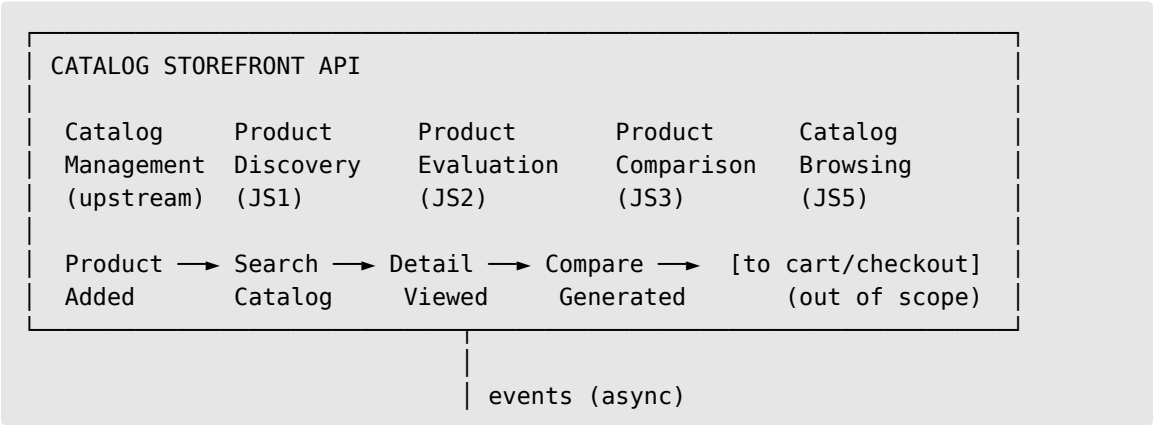
Changed → Update Partner Feed Cache **Policy:** Whenever Catalog Feed Generated → Log Feed Access for Partner Analytics

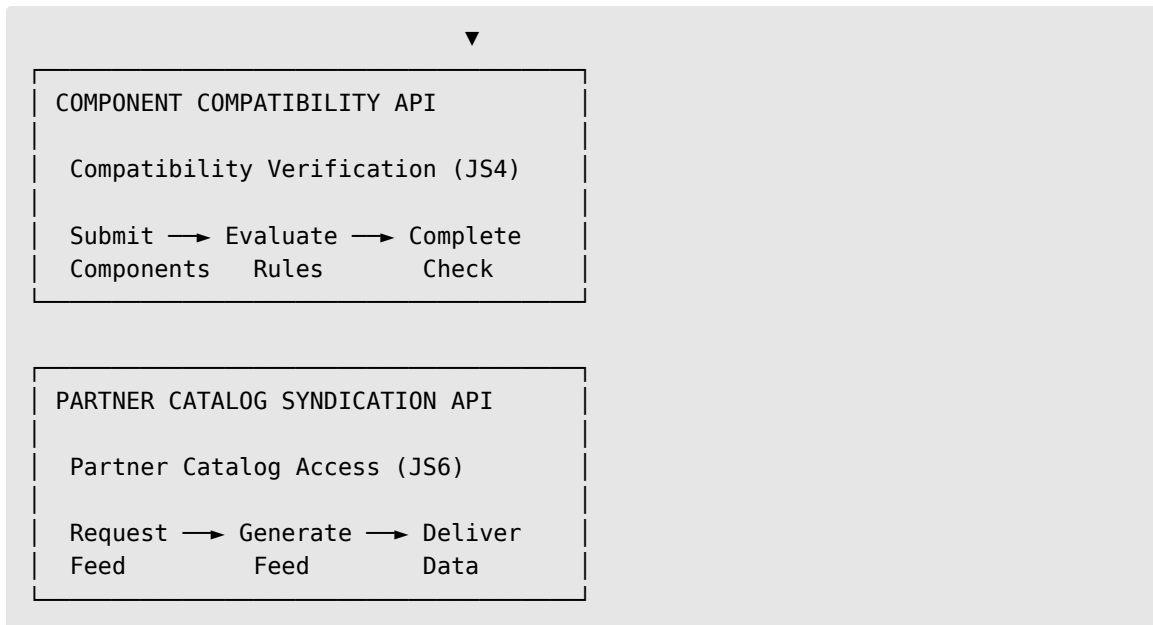
Pivotal Events Summary

Pivotal events mark significant transitions — where the domain shifts context, where a decision is made, or where integration happens.

#	Pivotal Event	Why It's Pivotal
1	Product Added to Catalog	Everything starts here. Triggers search indexing, compatibility cache updates, partner feed updates. Entry point for the entire storefront.
2	Product Discontinued	Cascading impact — must be removed from search, flagged in compatibility checks, updated in partner feeds. Affects all three bounded contexts.
3	Comparison Matrix Generated	Marks transition from “exploring” to “deciding.” Often the last step before a purchase decision or compatibility check.
4	Compatibility Check Completed	The moment of truth for a build. Pass → proceed to purchase. Fail → back to discovery/comparison. Key differentiator of the platform.
5	Catalog Feed Generated	Marks the boundary between our domain and partner domains. Data leaves our control. Projection/translation happens here.

Swimlanes (Bounded Context Alignment)





Hotspots 🔥 (Areas Needing Further Discussion)

#	Hotspot	Question
1	Compatibility Rule Management	Who creates/updates rules? Admin UI? Data import? ML-derived?
2	Search Index Technology	What powers the search? Elasticsearch? Algolia? Database full-text?
3	Product Data Ownership	Who is the source of truth for product data? Internal catalog admin? Supplier feeds?
4	Review Moderation	Are reviews submitted through this API or imported? Who moderates?
5	Analytics Events	Should viewed/searched/compared events feed into a separate analytics context?
6	Real-time vs. Cached Availability	Is availability checked live against inventory or served from a cache?
7	Partner Authentication	How do partners authenticate? API keys? OAuth? Rate limits per partner?

All Domain Events (Complete List)

#	Event	Aggregate	Context
1	Product Added to Catalog	Product	Catalog
2	Product Specifications Updated	Product	Catalog
3	Product Price Changed	Product	Catalog
4	Product Discontinued	Product	Catalog
5	Product Availability Changed	Product	Catalog
6	Product Categorized	Product	Catalog
7	Technical Document Uploaded	Technical Document	Catalog
8	Category Created	Category	Catalog
9	Category Hierarchy Updated	Category	Catalog
10	Catalog Searched	Search Index	Catalog
11	Search Filters Applied	Search Index	Catalog
12	Search Results Sorted	Search Index	Catalog
13	Search Page Retrieved	Search Index	Catalog
14	Suggestions Retrieved	Search Index	Catalog
15	Product Details Viewed	Product	Catalog
16	Technical Document Downloaded	Technical Document	Catalog
17	Product Pricing Viewed	Product	Catalog

#	Event	Aggregate	Context
18	Product Availability Checked	Product	Catalog
19	Product Reviews Viewed	Review	Catalog
20	Compatible Products Retrieved	Product	Catalog
21	Similar Products Retrieved	Product	Catalog
22	Frequently Bought Together Retrieved	Product	Catalog
23	Product Added to Comparison	Comparison	Catalog
24	Product Removed from Comparison	Comparison	Catalog
25	Comparison Matrix Generated	Comparison	Catalog
26	Category Tree Retrieved	Category	Catalog
27	Category Browsed	Category	Catalog
28	Category Facets Retrieved	Category	Catalog
29	Category Products Filtered	Category	Catalog
30	Compatibility Check Requested	Compatibility Check	Compatibility
31	Voltage Compatibility Evaluated	Compatibility Check	Compatibility
32	Current Rating Compatibility Evaluated	Compatibility Check	Compatibility
33	Physical Fit Compatibility Evaluated	Compatibility Check	Compatibility
34	Protocol Compatibility Evaluated	Compatibility Check	Compatibility

#	Event	Aggregate	Context
35	Weight/Thrust Compatibility Eval- uated	Compatibility Check	Compatibility
36	Compatibility Check Completed	Compatibility Check	Compatibility
37	Compatible Re- placements Re- trieved	Compatibility Check	Compatibility
38	Catalog Feed Gen- erated	Catalog Feed	Partner
39	Batch Availability Retrieved	Catalog Feed	Partner
40	Batch Pricing Re- trieved	Catalog Feed	Partner

Event Classification

Command Events (state-changing)

Events 1–9, 23–24, 30–36 — These change domain state

Query Events (observational)

Events 10–22, 25–29, 37–40 — These read state but may be tracked for analytics

Integration Events (cross-boundary)

Events 1–6 published by Catalog, consumed by Compatibility and Partner contexts

Pivotal Events (phase transitions)

Events 1, 4, 25, 36, 38 — Mark significant domain transitions